

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-11	Team No-1	Project Title-Smart Inventory & Warehouse Management System
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1. Abstract

(Summarize the project idea, problem background, and proposed solution. Not more than 250 words.)

The Smart Inventory & Warehouse Management System is a database-driven application designed to efficiently manage and monitor products, stock levels, suppliers, warehouses, and inventory transactions. The system provides a centralized platform for storing and maintaining inventory-related information, reducing the difficulties and errors associated with manual record keeping. The system maintains details such as product information, product categories, suppliers, warehouse locations, available stock, incoming and outgoing inventory, and transaction history. Whenever products are purchased, received, transferred, or dispatched, the corresponding stock information is updated in the database. The system can also identify products with low stock levels and help users determine when restocking is required. The project uses a Database Management System (DBMS) to organize data into related tables and establish relationships between entities such as products, suppliers, warehouses, and inventory transactions. SQL operations including insertion, updating, deletion, searching, joins, aggregate functions, views, and stored procedures can be used for efficient data management and retrieval. The proposed system improves inventory accuracy, minimizes data redundancy, simplifies warehouse operations, and provides quick access to important stock information. It demonstrates the practical application of DBMS concepts in developing a structured and reliable inventory and warehouse management solution.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090170	S Svava Haasini	
2520090227	B Sudharshini	
2520090157	Samyuktha Kandukuri	

3. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

Category	Recommended technology	Purpose
Database	MySQL	Stores products, suppliers, warehouses, stock, transactions.
Database tool	MySQL Workbench	Create tables, relationships, queries, views, procedures.
Backend	Python	Connects the application to the MySQL database.

Backend framework	Flask	Handles application logic and database
Frontend	HTML, CSS, JavaScript	Creates the user interface.
UI framework	Bootstrap	Makes the interface cleaner and responsive.
Code editor	Visual Studio Code	Develops and manages the project code.
Local platform	Localhost / Flask dev server	Runs and tests the application locally.

Guide Name and Signature (remarks if any) _____